Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Taygood

Selected Domain

Domain 2: Orbyte Smart SME Challenge (Alpha Red Solutions)

Selected SDG Track (1–17)

SDG 8: Decent Work and Economic Growth (targets 8.3 and 8.2)

Project / Solution Name

Kutip: cash-collection assistant for the Orbyte hub

Name (Leader)

Hansen Yap Hong San

Email & Phone (Leader)

yaphongsan@gmail.com · Phone: **TODO**

University Affiliation(s)

Taylor's University

Number of Team Members

TODO: number of members

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro and small businesses that sell to other businesses usually sell **on credit**: deliver now, get paid in 30 days. In practice they wait far longer. Experian reports SME payment delays of **64 days** on average (58 for large firms), and **76 days** in transport and logistics. Meanwhile salaries, rent and supplier bills fall due on fixed dates.

The owner of a 5-20 person firm (haulier, printer, wholesaler, catering supplier) carries this alone, mostly on WhatsApp and spreadsheets. The owner:

- finds out on payroll day that cash is short;
- keeps giving goods on credit to customers who already pay late, so the debt grows;
- chases customers at random, from memory, often late at night;
- matches "dah bank in boss" screenshots to the bank app by hand, and sometimes chases customers who have already paid.

Why it matters. Late payment is a cash-flow problem, not a profit problem, yet it is how otherwise healthy small firms miss salaries, pay their own suppliers late (passing the delay down the chain) or turn to costly short-term credit. Bank Negara names slower customer payments as one of three drivers of SME working-capital need. The burden falls on the businesses least able to absorb it, and on employees whose wages depend on it (SDG 8: decent work and the growth of micro and small enterprises). Existing reminder tools focus on chasing and target larger firms or specific accounting software; none warns the owner early or stops new credit to late payers.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Kutip is a cash-collection module for the Orbyte business hub. It promises only what the owner controls: **never be surprised on payroll day, and stop giving credit to customers who don't pay.**

One journey, end to end:

- **Trigger:** an 08:00 daily scan, a new credit order, or a customer reply.
- **Connected data:** invoices, bills, bank balance and statement lines, payment history (CSV standing in for Orbyte Connect).
- **Decision:** deterministic rules, each with a visible ID, forecast the cash gap and warn 10+ days ahead, hold new credit for customers 30+ days late, and rank who to chase by how much each invoice closes the gap.
- **Human approval:** the clerk approves gentle reminders; the owner approves firm reminders, credit decisions and deductions.
- **Outcome:** a WhatsApp reminder from a fixed BM/English/Chinese template, an invoice verified, the cash meter updated, and an append-only activity log.

AI is used in one place, where it adds value. Bukti reads messy replies and payment slips ("Bank in RM2,770, deduct RM330 for the 2 pallet tiles pecah") into structured fields using Claude Opus 5.5, a vision-capable LLM returning schema-validated JSON. Every field must quote the customer's exact words or it is dropped; fields below 85% confidence go to a person with one question; nothing is marked paid unless a matching bank line exists. AI never decides anything about money and never writes to customers.

Workshop skills we apply: GenAI APIs (image input and structured output, with a cached fallback); database design (an entity model for invoices, claims, promises and the audit log, money stored as integer sen, ready to move from the browser store to a hosted database in Phase 2); and cloud deployment (the Next.js app and its server-side AI route deploy to Vercel, with the API key kept in environment variables).

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS 4, lucide-react.
Data (Phase 1): synthetic seed in a Zustand store persisted to the browser; money as integer sen. **Phase 2:** hosted database (Convex or Postgres) behind the same engine.
Rules engine: pure TypeScript, deterministic; 97 Vitest tests replay every rule and the full demo script.
AI: Claude Opus 5.5 via the Anthropic API (official SDK); image + text in, JSON out, validated with Zod; cached results as fallback.
Messaging: WhatsApp click-to-chat (wa.me) links; a person presses send. **Hosting:** Vercel (deployment planned; runs locally today). **Code:** public GitHub repo.

Block diagram

- Browser UI (simulated Orbyte hub, Owner/Clerk roles) → Rules engine (forecast, credit check, chase plan, bank match) → Human approval → Outcome + activity log
- Customer reply + slip → /api/extract (Next.js server route) → Anthropic API → validated fields with exact quotes → Rules engine
- Synthetic invoices, bank CSV, messages and slips → Rules engine (stand-in for Orbyte Connect)

Sponsor-provided GenAI API or cloud platform: No. We use the sponsor's Orbyte concept (shared hub, roles, Connect, audit trail) as the frame for the module, simulated inside our app. No sponsor API, credits or cloud are used.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we researched Malaysian micro-SME pains (owner forums, Experian, Bank Negara, news) and found owners chasing payments on WhatsApp at night with payroll at risk.
Define: a customer can always refuse to pay, so we reframed the problem around what the owner controls: knowing early, the credit decision, who to chase, and correct records.
Ideate: we scored a shortlist of SME pains against the judging rubric and scanned competitors (Bukku, Paidnice, Peakflo) to find the gap: early warning and credit holds for micro firms that work on WhatsApp.
Prototype: a working module inside a simulated Orbyte hub, with synthetic data and a frozen demo clock so the whole journey runs live and repeatably.
Test: 97 automated tests replay every rule and the demo script; AI output is bounded by the exact-quote check and the bank match.
Ready for Phase 2: the app is one Next.js unit that deploys to Vercel. Next we interview 5-10 Malaysian B2B sellers, run timed first-use trials (goal: learned in under 10 minutes), measure extraction accuracy on 30-50 mixed-language messages, and move the engine onto a hosted database.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 8.3:** promote policies that support productive activities, decent job creation and entrepreneurship, and encourage the formalisation and growth of micro-, small- and medium-sized enterprises, including through access to financial services. **SDG 8.2:** achieve higher levels of economic productivity through technological upgrading and innovation. Supporting: SDG 9.3 (small enterprises' integration into value chains).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	• Malaysian SMEs wait **64 days** on average to be paid (58 days for large firms); transport and logistics **76 days** [1]. • Warning before a payroll shortfall today: **0 days**; owners find out on payroll day [2]. • Matching payment slips by hand: about **5 minutes per slip**, so about **10 hours a month** at 120 slips (team estimate, to be measured in interviews). • Invoices marked paid from a screenshot alone, without a bank check: not measured; we treat any above 0% as a risk. [1] Experian Malaysia, State of Credit 2025, 21 May 2025. [2] Owner post, forum.lowyat.net/topic/5206091: "Always 30 days became 60/90/120 days."

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	• Days of advance warning before a cash shortfall (days). • Share of invoices marked paid without a matching bank line (%). • Clerk time per payment slip matched (minutes per slip). • Exact-match accuracy of the amount read from mixed-language replies (%). • Time for a first-time user to learn the daily tasks (minutes).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	• **100%** of cash shortfalls in our test scenarios warned **at least 10 days** ahead (baseline: 0 days). • **0%** of invoices marked paid without a bank line (enforced by rule and by test). • **70% less** clerk time per slip: from about 5 minutes to 1.5 minutes or less, in timed trials with 3 or more first-time users. • **95% or higher** exact-match accuracy on amounts across a 30-50 message BM/English/Chinese test set. • First-time users approve a reminder and a payment match in **under 10 minutes** without training.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

None. All code, documents and research in this repository were created during the hackathon period (first commit 25 October 2026). No team member's earlier project code, design or research is used.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js 16 and React 19	Code (web framework and UI)	MIT
Tailwind CSS 4 and lucide-react	Code (styling and icons)	MIT; ISC
Zustand 5 and Zod 4	Code (state store and schema validation)	MIT
Anthropic TypeScript SDK 1.3	Code (API client)	Apache-2.0
Claude Opus 5.5 (Anthropic)	Model (pre-trained, called via API)	Anthropic Commercial Terms of Service

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

TODO: confirm whether any member has an employment, internship or consulting agreement that could claim this work. If none, write "Not applicable" and set the answer above to No.

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date] 25 October 2026